

Adventist University of Central Africa

Master's in Big Data Analytics

Chapter 3 — Linear Regression

Exercises 1, 6, 11

Introduction to Statistical Learning with Applications in R

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1 Conceptual Exercises

1.1 Exercise 1 — Null Hypotheses in Table 3.4

Question. Describe the null hypotheses to which the p-values given in Table 3.4 correspond. Explain what conclusions you can draw based on these p-values. Your explanation should be phrased in terms of **sales**, **TV**, **radio**, and **newspaper**, rather than in terms of the coefficients of the linear model.

Setting up the model. The Advertising data are fit with a single multiple regression model in which **sales** is regressed simultaneously on all three advertising budgets:

$$\text{sales} = \beta_0 + \beta_1 \cdot \text{TV} + \beta_2 \cdot \text{radio} + \beta_3 \cdot \text{newspaper} + \epsilon.$$

Table 3.4 (reproduced below from the text) reports the estimated coefficient, its standard error, the resulting *t*-statistic, and the p-value for each term:

Predictor	Coefficient	Std. error	<i>t</i> -statistic	p-value
Intercept	2.939	0.3119	9.42	< 0.0001
TV	0.046	0.0014	32.81	< 0.0001
radio	0.189	0.0086	21.89	< 0.0001
newspaper	−0.001	0.0059	−0.18	0.8599

Plain-language explanation. Each p-value in the table answers one narrow question: *if I already know how much was spent on the other two advertising channels, does knowing the spend on this particular channel still help me predict sales?* This is different from asking whether the channel is correlated with **sales** on its own — it is asking whether the channel carries **extra**, non-redundant information about **sales** once the other two channels are held fixed.

- For **TV**, the p-value is tiny (< 0.0001). This says: it is essentially impossible that TV spending has *no* effect on **sales** once radio and newspaper spending are held constant — the data would be extremely surprising under that assumption. So we conclude TV advertising is genuinely associated with higher sales, over and above whatever radio and newspaper are doing.
- For **radio**, the same reasoning applies: the p-value is tiny, so radio spending appears to have a real, additional effect on sales, holding TV and newspaper fixed.
- For **newspaper**, the p-value is large (0.8599). This says the data are entirely consistent with newspaper spending having *no* additional effect on sales once TV and radio are already accounted for. We do **not** have evidence that spending more on newspaper advertising increases sales, after controlling for the other two channels.

Formal statement. For each coefficient β_j , $j \in \{\text{TV}, \text{radio}, \text{newspaper}\}$, the p-value corresponds to testing

$$H_0 : \beta_j = 0 \quad \text{vs.} \quad H_a : \beta_j \neq 0,$$

holding the other two predictors fixed. At any conventional significance level (e.g. $\alpha = 0.05$), we **reject** H_0 for TV and radio (strong evidence of association with **sales**), and we **fail to reject** H_0

for newspaper (no evidence of association with **sales**, once TV and radio are already in the model).

A subtlety worth flagging. This does *not* mean newspaper spending is unrelated to sales in a simple, one-predictor regression — on its own it is mildly positively correlated with **sales**. The reason it drops out here is that newspaper spending is itself correlated with radio spending (markets that spend more on radio also tend to spend more on newspaper). Once radio is in the model, it “absorbs” the association that newspaper appeared to have on its own, and newspaper has nothing left to contribute. This is a classic confounding pattern, not a contradiction.

1.2 Exercise 6 — The Least Squares Line Passes Through $(\bar{x}, \bar{y})$

Question. Using (3.4), argue that in the case of simple linear regression, the least squares line always passes through the point $(\bar{x}, \bar{y})$.

Plain-language idea. The least squares line is chosen to minimize the total squared vertical distance between the data and the line. It turns out that the intercept term is *defined* in exactly the way needed to force the line through the “average point” of the data — the point whose coordinates are the mean of x and the mean of y . So this isn’t a coincidence that happens to hold for some datasets; it’s built into the formula for $\hat{\beta}_0$ itself, and therefore holds for *every* dataset.

Formal argument. Equation (3.4) gives the least squares estimates for simple linear regression as

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

The fitted regression line is $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$. Evaluate this line at $x = \bar{x}$:

$$\hat{y}\Big|_{x=\bar{x}} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x} = (\bar{y} - \hat{\beta}_1 \bar{x}) + \hat{\beta}_1 \bar{x} = \bar{y}.$$

The $\hat{\beta}_1 \bar{x}$ terms cancel exactly, leaving $\hat{y} = \bar{y}$ when $x = \bar{x}$. Hence the point $(\bar{x}, \bar{y})$ always satisfies the fitted equation — that is, it always lies on the least squares line, regardless of the particular data values. ■

Numerical check. The algebra above holds for *any* numeric data, so the following chunk checks it on one arbitrary, fixed data set. Because the data are fixed constants (no random number generation involved), the result is fully deterministic:

```
x <- c(1, 2, 3, 4, 5, 6, 7)
y <- c(2.1, 3.4, 5.0, 4.8, 6.3, 7.1, 7.9)

fit <- lm(y ~ x)

xbar <- mean(x)
ybar <- mean(y)
y_hat_at_xbar <- as.numeric(predict(fit, newdata = data.frame(x = xbar)))

data.frame(
  quantity = c("mean(x)", "mean(y)", "fitted value at mean(x)"),
  value     = c(xbar, ybar, y_hat_at_xbar)
)

##               quantity      value
## 1             mean(x) 4.000000
## 2             mean(y) 5.228571
## 3 fitted value at mean(x) 5.228571

isTRUE(all.equal(y_hat_at_xbar, ybar))

## [1] TRUE
```

The final line evaluates to **TRUE**: the fitted value at $\bar{x}$ equals $\bar{y}$ exactly (up to floating-point rounding), confirming the proof above.

2 Applied Exercises

2.1 Exercise 11 — Regression Through the Origin

Question setup. We generate a predictor x and a response y as follows, and investigate the t -statistic for $H_0 : \beta = 0$ in simple linear regression **without an intercept**.

```
set.seed(1)
x <- rnorm(100)
y <- 2 * x + rnorm(100)
```

2.1.1 (a) Regress y onto x , without an intercept

```
fit_yx <- lm(y ~ x + 0)
summary(fit_yx)

##
## Call:
## lm(formula = y ~ x + 0)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.9154 -0.6472 -0.1771  0.5056  2.3109
##
## Coefficients:
##      Estimate Std. Error t value Pr(>|t|)
## x    1.9939      0.1065   18.73  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9586 on 99 degrees of freedom
## Multiple R-squared:  0.7798, Adjusted R-squared:  0.7776
## F-statistic: 350.7 on 1 and 99 DF, p-value: < 2.2e-16
```

Comment. Since y was generated as (approximately) $2x$ plus noise, we would expect the slope estimate $\hat{\beta}$ to come out close to 2, with a standard error that is small relative to the estimate, and hence a large t -statistic and a very small p -value — strong evidence against $H_0 : \beta = 0$. Compare these expectations against the actual **Estimate**, **Std. Error**, **t value**, and **Pr(>|t|)** printed in the `summary()` output above: the fitted slope should sit close to 2, and the association should come through as highly statistically significant.

2.1.2 (b) Regress x onto y , without an intercept

```
fit_xy <- lm(x ~ y + 0)
summary(fit_xy)
```

```
##
## Call:
## lm(formula = x ~ y + 0)
```

```
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.8699 -0.2368  0.1030  0.2858  0.8938
##
## Coefficients:
##      Estimate Std. Error t value Pr(>|t|)
## y   0.39111    0.02089   18.73  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4246 on 99 degrees of freedom
## Multiple R-squared:  0.7798, Adjusted R-squared:  0.7776
## F-statistic: 350.7 on 1 and 99 DF,  p-value: < 2.2e-16
```

Comment. Now the roles are reversed: we predict x from y . Since $y \approx 2x$, inverting the relationship gives $x \approx y/2$, so we would expect the slope estimate here to be roughly 0.5 — noticeably different from the estimate in part (a). What is more interesting is what does *not* change: as parts (c)–(e) show, the t -statistic testing $H_0 : \beta = 0$ is identical in both directions, even though the coefficient estimates and their standard errors differ.

2.1.3 (c) Relationship between (a) and (b)

Both regressions are, in effect, testing the same underlying question: *is there a linear association between x and y ?* The coefficient estimates in (a) and (b) are not reciprocals of one another and have different standard errors, because they are measured on different scales (slope of y on x versus slope of x on y). However, the strength of evidence for association — captured by the t -statistic and p-value — turns out to be exactly the same in both directions. Parts (d) and (e) show why this must be true algebraically, and the R output above/below confirms it numerically.

2.1.4 (d) Algebraic derivation of the t -statistic formula

For regression through the origin, (3.38) gives

$$\hat{\beta} = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i'=1}^n x_{i'}^2},$$

and the exercise gives the standard error as

$$SE(\hat{\beta}) = \sqrt{\frac{\sum_{i=1}^n (y_i - x_i \hat{\beta})^2}{(n-1) \sum_{i'=1}^n x_{i'}^2}}.$$

We want to show that $t = \hat{\beta}/SE(\hat{\beta})$ can be rewritten as

$$t = \frac{\sqrt{n-1} \sum_{i=1}^n x_i y_i}{\sqrt{(\sum_{i=1}^n x_i^2) (\sum_{i'=1}^n y_{i'}^2) - (\sum_{i'=1}^n x_{i'} y_{i'})^2}}.$$

Step 1 — expand the residual sum of squares. Write $S_{xy} = \sum_i x_i y_i$, $S_{xx} = \sum_i x_i^2$, $S_{yy} = \sum_i y_i^2$. Expanding the square:

$$\sum_i (y_i - x_i \hat{\beta})^2 = \sum_i y_i^2 - 2\hat{\beta} \sum_i x_i y_i + \hat{\beta}^2 \sum_i x_i^2 = S_{yy} - 2\hat{\beta} S_{xy} + \hat{\beta}^2 S_{xx}.$$

Step 2 — substitute $\hat{\beta} = S_{xy}/S_{xx}$.

$$2\hat{\beta} S_{xy} = \frac{2S_{xy}^2}{S_{xx}}, \quad \hat{\beta}^2 S_{xx} = \frac{S_{xy}^2}{S_{xx}}.$$

So

$$\sum_i (y_i - x_i \hat{\beta})^2 = S_{yy} - \frac{2S_{xy}^2}{S_{xx}} + \frac{S_{xy}^2}{S_{xx}} = S_{yy} - \frac{S_{xy}^2}{S_{xx}} = \frac{S_{xx} S_{yy} - S_{xy}^2}{S_{xx}}.$$

Step 3 — assemble the t -statistic.

$$t = \frac{\hat{\beta}}{SE(\hat{\beta})} = \frac{S_{xy}/S_{xx}}{\sqrt{\frac{1}{(n-1)S_{xx}} \left(\frac{S_{xx} S_{yy} - S_{xy}^2}{S_{xx}} \right)}} = \frac{S_{xy}/S_{xx}}{\sqrt{\frac{S_{xx} S_{yy} - S_{xy}^2}{(n-1)S_{xx}^2}}}.$$

Since $\sqrt{1/S_{xx}^2} = 1/S_{xx}$ (as $S_{xx} > 0$), the S_{xx} in the denominator of $\hat{\beta}$ cancels with the S_{xx} pulled out of the square root:

$$t = \frac{S_{xy}/S_{xx}}{\frac{1}{S_{xx}} \sqrt{\frac{S_{xx} S_{yy} - S_{xy}^2}{n-1}}} = \frac{S_{xy}}{\sqrt{\frac{S_{xx} S_{yy} - S_{xy}^2}{n-1}}} = \frac{\sqrt{n-1} S_{xy}}{\sqrt{S_{xx} S_{yy} - S_{xy}^2}},$$

which is exactly the target formula, written back in full as

$$t = \frac{\sqrt{n-1} \sum_i x_i y_i}{\sqrt{(\sum_i x_i^2)(\sum_i y_i^2) - (\sum_i x_i y_i)^2}}. \quad \blacksquare$$

Numerical confirmation in R. The chunk below computes t three independent ways — by hand from $\hat{\beta}$ and $SE(\hat{\beta})$, from the closed-form formula just derived, and by reading it off `lm()`'s own summary — and checks that all three agree:

```
n <- length(x)

beta_hat <- sum(x * y) / sum(x^2)
se_beta <- sqrt(sum((y - x * beta_hat)^2) / ((n - 1) * sum(x^2)))
t_manual <- beta_hat / se_beta

t_formula <- (sqrt(n - 1) * sum(x * y)) /
```

```

sqrt(sum(x^2) * sum(y^2) - (sum(x * y))^2)

t_lm <- summary(fit_yx)$coefficients["x", "t value"]

data.frame(
  method = c("by hand (beta_hat / SE)", "closed-form formula", "lm() summary"),
  t_stat = c(t_manual, t_formula, as.numeric(t_lm))
)

##               method    t_stat
## 1 by hand (beta_hat / SE) 18.72593
## 2   closed-form formula 18.72593
## 3           lm() summary 18.72593

isTRUE(all.equal(t_manual, t_formula)) &&
isTRUE(all.equal(t_manual, as.numeric(t_lm)))

## [1] TRUE

```

The final line should evaluate to TRUE, confirming that the algebra above matches both a from-scratch numerical computation and what `lm()` reports.

2.1.5 (e) The t -statistic is the same for y on x as for x on y

Look at the closed-form expression from part (d):

$$t = \frac{\sqrt{n-1} \sum_i x_i y_i}{\sqrt{(\sum_i x_i^2) (\sum_i y_i^2) - (\sum_i x_i y_i)^2}}.$$

This formula treats x and y almost, but not quite, symmetrically — the only way x and y enter is through the three summary quantities $S_{xy} = \sum_i x_i y_i$, $S_{xx} = \sum_i x_i^2$, and $S_{yy} = \sum_i y_i^2$. If instead we regress x onto y (without an intercept), the same derivation applies with the roles of x and y swapped throughout, giving

$$t' = \frac{\sqrt{n-1} \sum_i y_i x_i}{\sqrt{(\sum_i y_i^2) (\sum_i x_i^2) - (\sum_i y_i x_i)^2}}.$$

But $\sum_i x_i y_i = \sum_i y_i x_i$ (multiplication of real numbers is commutative), and the product $(\sum_i y_i^2) (\sum_i x_i^2)$ is the same regardless of the order in which the two sums are multiplied. So the numerator and denominator of t' are literally the same expressions as in t , just written in a different order: $t' = t$. This is exactly the pattern already glimpsed numerically in parts (a)–(b): the coefficient estimates differ, but the t -statistic for $H_0 : \beta = 0$ does not.

2.1.6 (f) With an intercept: same check in R

```

fit_yx_int <- lm(y ~ x)
fit_xy_int <- lm(x ~ y)

```

```

t_yx_int <- summary(fit_yx_int)$coefficients["x", "t value"]
t_xy_int <- summary(fit_xy_int)$coefficients["y", "t value"]

data.frame(
  regression = c("y ~ x", "x ~ y"),
  t_stat      = c(as.numeric(t_yx_int), as.numeric(t_xy_int))
)

##   regression  t_stat
## 1      y ~ x 18.5556
## 2      x ~ y 18.5556

isTRUE(all.equal(as.numeric(t_yx_int), as.numeric(t_xy_int)))

## [1] TRUE

```

The final line should again evaluate to **TRUE**. This is not a coincidence special to regression through the origin: for simple linear regression **with** an intercept, it is a standard result that the t -statistic for testing $H_0 : \beta_1 = 0$ satisfies

$$t^2 = \frac{(n-2) r_{xy}^2}{1 - r_{xy}^2},$$

where r_{xy} is the sample correlation between x and y . Because correlation is symmetric ($r_{xy} = r_{yx}$), this expression is identical whether we regress y on x or x on y — so the two regressions must again produce the same t -statistic, exactly as the R output above confirms.